

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	31 January 2025
Team ID	LTVIP2025TMID45199
Project Name	Enchanted Wings: Marvels of Butterfly Species
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Reference: <https://www.mural.co/templates/brainstorm-and-idea-prioritization>

Purpose

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your brainstorming sessions to shape your butterfly classification system. Even if your team is remote, this format ensures structured, inclusive ideation that drives real innovation.

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Brainstorm & idea prioritization

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

⌚ 10 minutes to prepare
👤 1 hour to collaborate
👥 2-8 people recommended

➡ Before you collaborate
A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

⌚ 10 minutes

- A Team gathering**
Define who should participate in the session and send an invite. Share relevant information or pre-work ahead.
- B Set the goal**
Think about the problem you'll be focusing on solving in the brainstorming session.
- C Learn how to use the facilitation tools**
Use the Facilitation Superpowers to run a happy and productive session.

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1 Define your problem statement
What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

⌚ 5 minutes

PROBLEM

How might we [your problem statement]?

Key rules of brainstorming

To run a smooth and productive session

- 🗨️ Stay in topic.
- 💡 Encourage wild ideas.
- 🙊 Defer judgment.
- 👂 Listen to others.
- 🗣️ Go for volume.
- 👁️ If possible, be visual.

Problem Statement:

Develop an intelligent butterfly image classification model using transfer learning to identify species accurately. This system supports biodiversity monitoring, ecological research, and educational engagement.

Step-2: Brainstorm, Idea Listing and Grouping

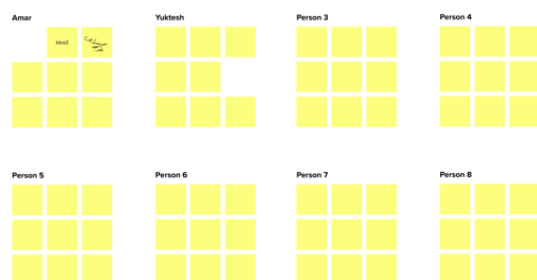
2

Brainstorm

Write down any ideas that come to mind that address your problem statement.

🕒 10 minutes

TIP
You can select a sticky note and hit the pencil icon to start drawing!



3

Group Ideas

Take turns sharing your ideas while clustering similar or related notes as you go. In the last 10 minutes, give each cluster a sentence-like label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub-groups.

🕒 20 minutes

Person 4

TIP
Add customizable tags to sticky notes to make it easier to find, browse, organize, and categorize important ideas as themes within your mural.

💡 Step-2: Brainstorm, Idea Listing, and Grouping

Idea	Description	Category
Transfer Learning	Use VGG16/ResNet-based CNNs to leverage existing models for improved accuracy and training speed	Technical Strategy
Grouped Dataset Preparation	Prepare train/validation/test sets from the 6499-image, 75-class butterfly dataset	Data Preparation
Real-Time Image Classification	Deploy model via web app (e.g., Streamlit) for live species detection	End-user Experience

Visuals and Results in Report	Capture training graphs, confusion matrix, and predictions for final report	Documentation & Impact
Integration with Citizen Science	Allow users to contribute photos and receive ID feedback	Engagement & Learning
Ecosystem Application	Showcase use in biodiversity tracking and ecological studies	Real-World Application

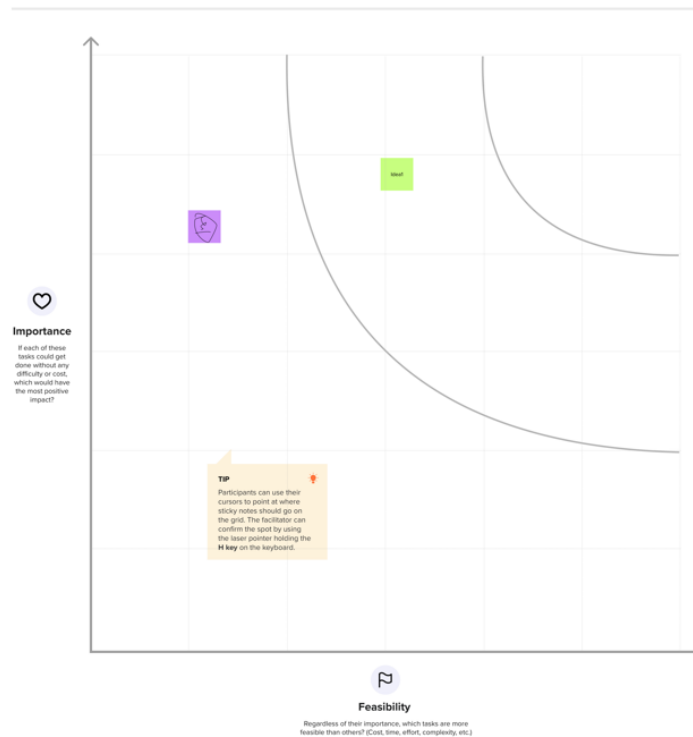
Step-3: Idea Prioritization

4

Prioritize

Your team should all be on the same page about what's important moving forward. Place your ideas on this grid to determine which ideas are important and which are feasible.

20 minutes



Step-3: Idea Prioritization

Priority

Idea

Why It's Important

High	Transfer Learning	It forms the backbone of your accurate species recognition system
High	Dataset Preparation	Properly organized data improves training results significantly
High	Use Case Mapping	Establishes direct value in conservation and biodiversity research
Medium	Streamlit Real-Time Prediction	Increases accessibility and encourages user interaction
Medium	Citizen Science Integration	Adds public value and broadens data gathering potential
Low	Screenshots for Report	Important for presentation, but doesn't influence model performance